

Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science and Engineering		
Programme Name	B.E (Computer Science & Engineering)		
Course Name	Algorithm Design & Implementation	Session	July-Dec 2026
Course Code	24CSE0317	Semester/Batch	5 th /2024
L-T-P (Per Week)	2-0-4	Course Credits	04
Pre-requisite	Fundamentals of C Programming (23CS003), Data Structures (23CS004)	NHEQF Level ¹	5.5
Course Coordinator	Dr. Astha Gupta	SDG Number ⁴	4,8,9

1. Objectives of the Course

The scope of the course is to provides the foundation for understanding the key aspects of java programming and implementation obtaining a theoretical understanding of advance programming concepts. The objectives of the course are:

- to build an understanding of analysing and evaluating algorithm efficiency.
- to inculcate the skill in students to implement and optimize data structures.
- to develop and apply advanced sorting and searching algorithms.
- to solve complex problems using greedy algorithms, backtracking, and dynamic programming.

2. Course Learning Outcomes

After completion of the course, the student should be able to:

	Course Learning Outcome	*POs	NHEQF Level Descriptor ²	Sessions
CLO01	Understand and apply the concept of algorithm complexity	PO1, PO2, PO3, PO4, PO5, PO11	Q1, Q2	16
CLO02	Proficiency in implementing hash tables, heaps, priority queues	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q2, Q3	16
CLO03	Implement the advanced sorting algorithms and apply appropriate algorithm for a particular problem	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q2, Q3, Q4	16
CLO04	Apply the concepts and theories of different algorithmic strategies	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q1, Q2, Q4	18
CLO05 (Only for lab components)	Illustrate the concept of Array, String, Stack, Queue, Hashing, Mapping, Recursion, Heap, Tree and Graph data structures.	PO1, PO2, PO3, PO4, PO5, PO9, PO11	Q1, Q2, Q3, Q4, Q5, Q6	24
Total Contact Hours				90

CLO-PO Mapping

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 01	PSO 02	Type of Assessment's ³
CLO01	H	H	L	M	M	-	-	-	-	-	M	H	M	Formative, Summative
CLO02	H	M	M	L	H	-	-	L	-	-	M	H	H	Formative, Summative
CLO03	H	H	H	M	H	-	-	L	-	-	M	H	H	Formative, Summative
CLO04	H	H	H	M	M	-	-	L	-	-	H	H	M	Formative, Summative
CLO05	H	M	L	L	M	-	-	-	L	-	L	H	M	Formative, Summative

H=High, M=Medium, L=Low

¹ National Higher Education Qualification Framework Level, Refer to annexure

² NHEQF Level Descriptor, Refer to Annexure & [Learning outcomes descriptors for qualification for all levels on the NHEQF](#)

³Types of Assessments can be referred from Type of Assessments. Refer to Annexure.

⁴For SDG Mapping with Courses, PI refer [SDG Mapping policy for Courses](#)

3. Recommended Books:

- B01:** “Classic Data Structures”, Samanta and Debasis, 2nd edition, PHI publishers.
B02: “Fundamentals of Data Structures”, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
B03: "Design and Analysis of Algorithms" by S. Sridhar, Oxford University Press, 2014.
B04: “Data Structures with C (Schaum's Outline Series)”, Seymour Lipschutz, 1st edition, McGraw Hill Education.
B05: “Algorithms, Data Structures, and Problem Solving with C++”, Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company.

E-Resources:

- <https://library.chitkara.edu.in/subscribed-books.php>

4. Other readings and relevant websites:

S. No.	Link of Journals, Magazines, websites and Research Papers
1.	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart
2.	https://www.geeksforgeeks.org/distributing-m-items-circle-size-n-starting-k-th-position/
3.	https://www.geeksforgeeks.org/open-addressing-collision-handling-technique-in-hashing/
4.	https://www.geeksforgeeks.org/find-the-winner-of-the-game-2/
5.	https://www.geeksforgeeks.org/priority-queue-using-linked-list/
6.	https://www.youtube.com/watch?v=HqPJF2L5h9U

5. Recommended Tools and Platforms

Test Pad

6. Course Plan (Theory + Lab Plan):

Theory Plan

Lecture Number	Topics
1-2	Unit-1: Overview of ADS, Time and Space Complexity – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.
3-5	Unit-2: Arrays & Strings – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro, String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems.
6-10	Unit-3: Greedy Algorithms – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems, Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.
11-15	Unit-4: Stack & Queue – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.
16-19	Unit-5: Hashing & Mapping – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints, String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.
20-22	Unit-6: Recursion & Backtracking – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems, Backtracking with Pruning, Optimization, Backtracking + DP (Memoization) Intro, Practice Contest 8: 3 backtracking problems, Practice Contest 9: 3 mixed recursion/backtracking problems.
23-25	Unit-7: Trees – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil, Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.
26-28	Unit-8: Heap – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.

29-30	Unit-9: Graphs – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.
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Lab Plan

Lab Number	Practical
1-6	Unit-1: Introduction to Time and Space Complexity: - Find the prime factors of a given number. - Compute GCD of two numbers using Euclidean Algorithm. - Distribute N items in a circle and find who gets the last one. - Solve a problem using brute force. - Give one example of a randomized algorithm. - Use divide and conquer to sort an array.
7-17	Unit-2: Arrays & Strings - Find a pair with sum K using two pointers. - Find the maximum sum subarray of size k using the sliding window technique. - Rotate an array using the reversal algorithm. - Check whether two strings are anagrams. - Find the Longest Common Subsequence (LCS). - Find the Longest Increasing Subsequence (LIS) using Dynamic Programming.
18-25	Unit-3: Greedy Algorithms - Apply the greedy approach to interval scheduling. - Solve the fractional knapsack problem. - Schedule jobs with deadlines to maximize profit. - Solve the activity selection problem using the greedy method.
26-32	Unit-4: Stack & Queue - Implement stack operations using an array. - Check balanced parentheses using a stack. - Implement circular queue operations.
33-38	Unit-5: Hashing & Mapping - Solve the "Noise in the Library" problem using hashing. - Use hashing to balance a scale with given weights. - Find the winner in a voting system using hashing.
39-45	Unit-6: Recursion & Backtracking - Solve the Tower of Hanoi using recursion. - Generate permutations of a string using backtracking. - Move a robot into a grid using backtracking. - Solve Sudoku using backtracking. - Find all paths for Rat in a Maze. - Print all binary strings of n bits. - Solve the N-Queens problem using backtracking. Dynamic Programming Applications: - Solve the 0/1 Knapsack problem using Dynamic Programming. - Count the number of ways to cover a distance of N using 1, 2, or 3 steps. - Solve Matrix Chain Multiplication using Dynamic Programming.
46-51	Unit-7: Trees - Perform preorder, inorder, and postorder traversals of a binary tree. - Find the height of a binary tree. - Find the Lowest Common Ancestor (LCA) of two nodes in a Binary Search Tree. - Find the kth smallest element in a Binary Search Tree.
52-56	Unit-8: Heap - Find the maximum/minimum element in a continuous stream of data. - Sort an array using Heap Sort. - Check whether a given tree is a Max-Heap.
57-60	Unit-9: Graphs - Find the cycle in an undirected graph. - Find the minimum number of edges in a path of a graph. - Find a path in a directed graph. - Find the number of islands. - Find the shortest path using Dijkstra's algorithm. - Solve a graph with negative edge weights using the Bellman-Ford algorithm.

7. Delivery/Instructional Resources

Plan:

Lecture No.	Topics	CLO	Book No, CH No, Page No	TLM ³	ALM ⁴	Web References	Audio-Video
1-2	Unit-1: Overview of ADS, Time and Space Complexity – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.	1	BO1-CH 1, BO4- CH 1, BO5- CH 2	Lecture Discussion	Quiz, Challenge, Assignment	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart	https://www.youtube.com/watch?v=HfIH3czXc-8
3-5	Unit-2: Arrays & Strings – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro, String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems.	3,4	BO2- CH 2, BO3- CH 2, BO5- CH 3	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/	https://www.youtube.com/watch?v=RBSGKIAvoIM
6-10	Unit-3: Greedy Algorithms – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems, Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.	2,4	BO4- CH 8, BO5- CH 10	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/greedy-algorithms/	https://www.youtube.com/watch?v=ARvQcqJ_-NY
11-15	Unit-4: Stack & Queue – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.	3	BO1- CH 3 & 4, BO2- CH 3 & 4, BO3- CH 5 & 6	Lecture Discussion	Quiz, Challenge, Assignment		https://www.youtube.com/watch?v=wjI1WNcIntg
16-19	Unit-5: Hashing & Mapping – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints, String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.	2,4	BO1- CH 9, BO5- CH 5	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/string/string-hashing.html	https://www.youtube.com/watch?v=shs0KM3wKv8

³ Teaching Learning Methods, Refer to Annexure

⁴ Active Learning Methods

20-22	Unit-6: Recursion & Backtracking – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems, Backtracking with Pruning, Optimization, Backtracking + DP (Memoization) Intro, Practice Contest 8: 3 backtracking problems, Practice Contest 9: 3 mixed recursion/backtracking problems.	4	BO4- CH 4 & 5, BO5- CH 10	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/algebra/all-submasks.html	https://www.youtube.com/watch?v=DKCbsiDBN6c
23-25	Unit-7: Trees – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil, Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.	2,4	BO1- CH 6 & 7, BO2- CH 5, BO5- CH 4	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/graph/lca.html	https://www.youtube.com/watch?v=fAAZixBzIAI
26-28	Unit-8: Heap – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.	2,3	BO1- CH 8, BO2- CH 8, BO5- CH 6	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/algebra/all-submasks.html	https://www.youtube.com/watch?v=DKCbsiDBN6c
29-30	Unit-9: Graphs – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.	4	BO1- CH 10, BO4- CH 9, BO5- CH 9	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/backtracking-algorithms/	https://www.youtube.com/watch?v=QXkY2DOUCWM

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-6	Unit-1: Introduction to Time and Space Complexity: - Find the prime factors of a given number. - Compute GCD of two numbers using Euclidean Algorithm. - Distribute N items in a circle and find who gets the last one. - Solve a problem using brute force. - Give one example of a randomized algorithm. - Use divide and conquer to sort an array.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=HfIH3czXc-8 (Abdul Bari – Time Complexity & Algorithm Analysis)

7-17	Unit-2: Arrays & Strings - Find a pair with sum K using two pointers. - Find the maximum sum subarray of size k using the sliding window technique. - Rotate an array using the reversal algorithm. - Check whether two strings are anagrams. - Find the Longest Common Subsequence (LCS). - Find the Longest Increasing Subsequence (LIS) using Dynamic Programming.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=RBSGKlAvoiM (freeCodeCamp – Data Structures and Algorithms Full Course)
18-25	Unit-3: Greedy Algorithms - Apply the greedy approach to interval scheduling. - Solve the fractional knapsack problem. - Schedule jobs with deadlines to maximize profit. - Solve the activity selection problem using the greedy method.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=ARvQcQJ_-NY (Abdul Bari – Greedy Method)
26-32	Unit-4: Stack & Queue - Implement stack operations using an array. - Check balanced parentheses using a stack. - Implement circular queue operations.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=wjI1WNcIntg (freeCodeCamp – Stack Data Structure)
33-38	Unit-5: Hashing & Mapping - Solve the "Noise in the Library" problem using hashing. - Use hashing to balance a scale with given weights. - Find the winner in a voting system using hashing.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=shs0KM3wKv8 (Abdul Bari – Hashing)
39-45	Unit-6: Recursion & Backtracking - Solve the Tower of Hanoi using recursion. - Generate permutations of a string using backtracking. - Move a robot into a grid using backtracking. - Solve Sudoku using backtracking. - Find all paths for Rat in a Maze. - Print all binary strings of n bits. - Solve the N-Queens problem using backtracking. Dynamic Programming Applications: - Solve the 0/1 Knapsack problem using Dynamic Programming. - Count the number of ways to cover a distance of N using 1, 2, or 3 steps. - Solve Matrix Chain Multiplication using Dynamic Programming.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=QXkY2DOUCWM (Backtracking & Recursion Playlist – N-Queens, Sudoku, Rat in a Maze)
46-51	Unit-7: Trees - Perform preorder, inorder, and postorder traversals of a binary tree. - Find the height of a binary tree. - Find the Lowest Common Ancestor (LCA) of two nodes in a Binary Search Tree. - Find the kth smallest element in a Binary Search Tree.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=fAAZixBzIAI (William Fiset – Binary Trees & BST)

52-56	Unit-8: Heap 1. Find the maximum/minimum element in a continuous stream of data. 2. Sort an array using Heap Sort. 3. Check whether a given tree is a Max-Heap.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=t0Cq6tVNRBA (Abdul Bari – Heap Data Structure)
57-60	Unit-9: Graphs 1. Find the cycle in an undirected graph. 2. Find the minimum number of edges in a path of a graph. 3. Find a path in a directed graph. 4. Find the number of islands. 5. Find the shortest path using Dijkstra's algorithm. 6. Solve a graph with negative edge weights using the Bellman-Ford algorithm.	6	Demonstration method using a simulation or a tool	Lab Challenge	NA	https://www.youtube.com/watch?v=09_LIHjoEiY (William Fiset – Graph Theory Playlist)

8. Remedial Classes⁵

Slow Learners	Average Learners	Fast Learners
- Remedial Classes on Saturdays - Encouragement for improvement using Peer Tutoring - Use of Audio and Visual Materials - Use of Real-Life Examples	- Workshops - Formative Exercises used to highlight concepts and notions - E-notes and E-exercises to read ahead of the pedagogic material.	- Engaging students to hold hands of slow learners by creating a Peer Tutoring Group - Design solutions for complex problems - Design Solutions for complex problems - Presentation on topics beyond those covered in CHO

9. Self-Learning⁶

Assignments to promote self-learning, a survey of contents from multiple sources.

S. No.	Topics	CLO	ALM	References/MOOCs
1	Algorithm Analysis & Complexity	1	Group Discussion / Coding Practice	https://nptel.ac.in/courses/106105164
2	Arrays, Strings & Searching Techniques	3	Group Discussion / Coding Practice	https://www.coursera.org/learn/cpsc-8400-design-and-analysis-of-algorithms
3	Stack, Queue & Priority Queue Applications	2	Group Discussion / Coding Practice	https://nptel.ac.in/courses/106106145
4	Hashing & String-Matching Algorithms	2	Group Discussion / Coding Practice	https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/
5	Heaps & Priority Queues	2	Group Discussion / Coding Practice	https://www.coursera.org/learn/algorithms-part1
6	Trees, BST & Trie	2	Group Discussion / Coding Practice	https://www.coursera.org/learn/algorithms-part1

10. Delivery Details of Content Beyond Syllabus⁷

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1.	Advanced Data Structures: Segment Tree & Fenwick Tree	2	PO1, PO2	Programming Assignment	https://cp-algorithms.com/data_structures/segment_tree.html
2.	String Algorithms: Rabin-Karp, Aho-Corasick	2	PO1, PO2, PO3	Problem-Based Learning	https://cp-algorithms.com/string/

3.	Dynamic Programming Optimization (Bitmask DP, State DP)	4	PO1, PO2, PO3	Problem-Based Learning	https://cp-algorithms.com/dynamic_programming/
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11. Evaluation Scheme & Components:

Assessment Type ⁸	Evaluation Component ⁹	Type of Component ¹⁰	No. of Assessments ¹¹	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative, Summative	Test Pad Coding Series	Component 1	01	10%	10	Online/ Off Campus	1-5
Formative, Summative	Formal Assessment (FAs)		05*	10%	10	Online/ On Campus	1-5
Formative, Summative	Sessional Tests (STs)	Component 2	02*	30%	30	Online/ On Campus	1-5
Summative	End Term Examination	Component 3	01	50%	50	Online/ On Campus	1-5

* Out of 05 FAs, best 4 FA for final marks evaluation of FAs will be considered.

* Out of 02 STs, best 1 ST for final marks evaluation of STs will be considered.

* Makeup Examination will compensate for either ST-1 or ST-2 (Only for genuine cases, based on the Dean's approval and the exam will be based on the full syllabus).

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

⁵ Refer to Annexure

⁶ Refer to Annexure

⁷ Refer to Annexure

⁸ Refer to [Annexure 2 of NCrf](#)

⁹ Refer to Annexure

¹⁰Refer to Annexure

¹¹ Refer to Annexure

12. Syllabus of the Course:

S.No.	Subject		
		No. of Sessions	Weightage (%)
1	Unit-1: Overview of ADS, Time and Space Complexity – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.	6	7%
2	Unit-2: Arrays & Strings – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro. String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems	15	17%
3	Unit-3: Greedy Algorithms – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems. Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.	12	13%
4	Unit-4: Stack & Queue – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.	10	11%
5	Unit-5: Hashing & Mapping – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints. String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.	11	12%
ST-1 (Covering 60% Syllabus)			
6	Unit-6: Recursion & Backtracking – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems.	11	12%
7	Unit-7: Trees – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil. Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.	12	13%
8	Unit-8: Heap – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.	6	7%
9	Unit-9: Graphs – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.	7	8%
ST-2 (Covering 40% Syllabus)			
End Term 100% Syllabus			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Astha Gupta	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date	23-06-2026	